

UPPER MODERATE DEVIATION PROBABILITIES FOR THE MAXIMUM OF A BRANCHING RANDOM WALK

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- 1 Branching random walk
- 2 Asymptotic behavior of the maximum
- 3 Proof of the main result: first ideas
- 4 Applications

Branching random walk (BRW)

Definition:

- At time $n = 0$, a single particle at position $x = 0$.
- At time $n = 1$, this particle dies and gives birth to children, scattered in $\mathbb{R}$ according to a point process $\mathcal{L}$.
- Each child then restarts the process from its position, independently of the others, according to independent copies of $\mathcal{L}$.

Motivations:

- Biology (model of a population invading its environment).
- Statistical physics (directed polymers, spin glasses...).
- Other related models (BBM, log-correlated fields...).

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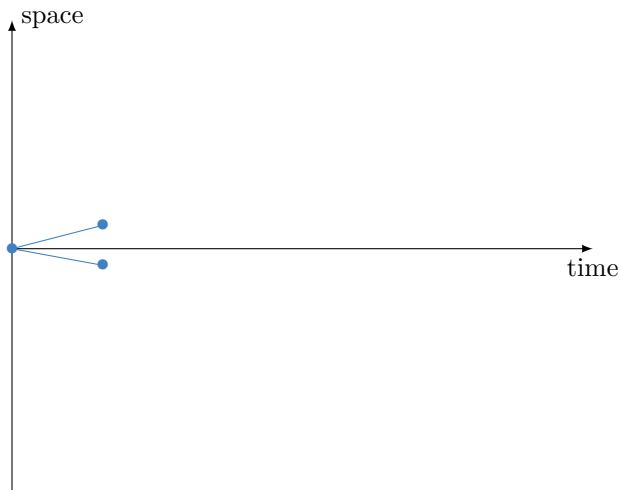


Figure: Realization of a branching random walk until time $n = 6$.

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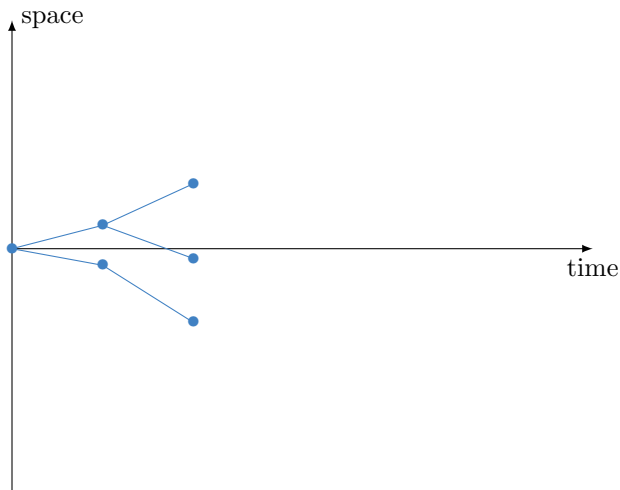


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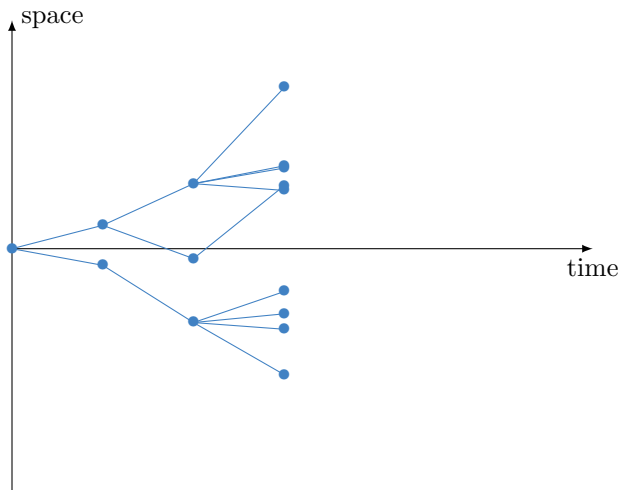


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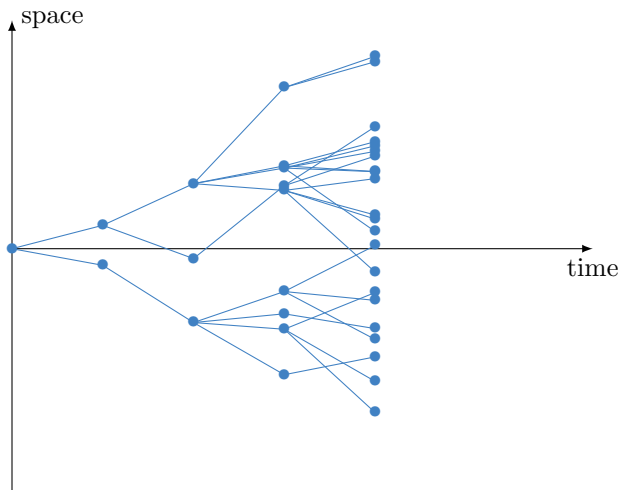


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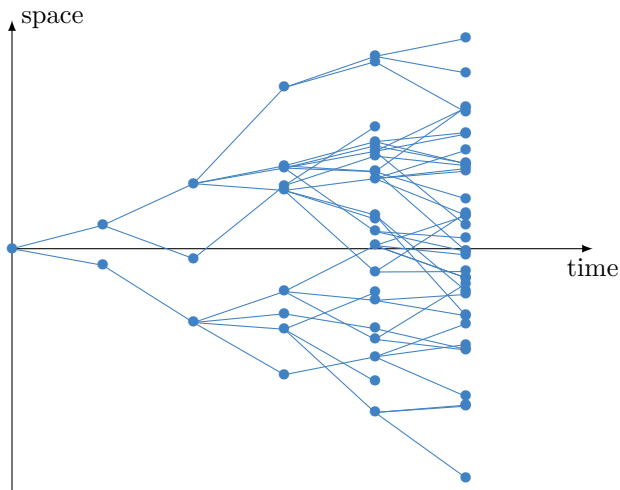


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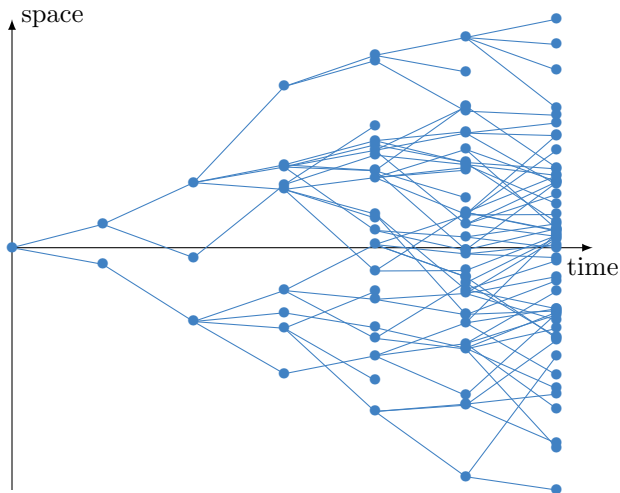


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The lattice and non-lattice cases

The BRW is:

- (a, h) -lattice if $h > 0$ is the largest number such that $\text{Supp } \mathcal{L} \subset a + h\mathbb{Z}$,
- non-lattice if such a and h do not exist.

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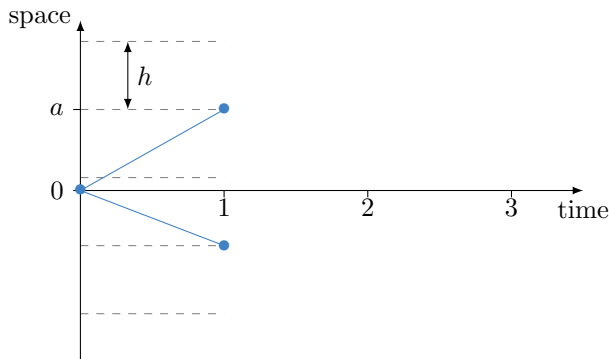


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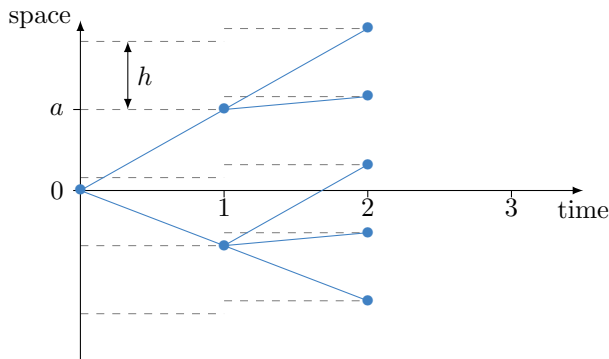


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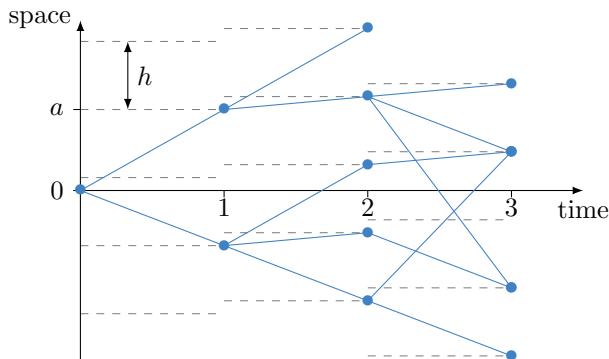


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Notations and assumptions

Notation:

- $V(u)$: position of particle u .
- $|u|$: generation of particle u .
- $M_n = \max_{|u|=n} V(u)$.

Assumptions:

(A1) Supercriticality: $\mathbb{E}[\sum_{|u|=1} 1] \in (1, \infty]$.

(A2) Boundary setting: $\mathbb{E}[\sum_{|u|=1} e^{V(u)}] = 1$ and $\mathbb{E}[\sum_{|u|=1} V(u)e^{V(u)}] = 0$.

(A3) Usual integrability conditions: same as [Aïdékon, 2013].

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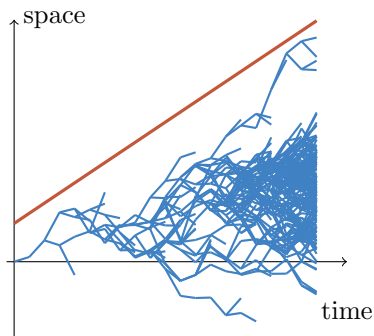
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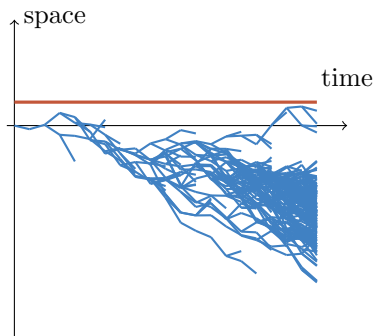
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The boundary case

We can assume (A2) without (too much) loss of generality [Jaffuel, 2009].



(a) A branching random walk



(b) Reduction to the boundary case

Figure: A branching random walk on $\mathbb{R}$ and its normalization to the boundary case. (Figure by [Mallein, 2016])

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Maximum of the BRW: typical fluctuations

Derivative martingale: $D_n = -\sum_{|u|=n} V(u)e^{V(u)}.$

Theorem ([Aïdékon, 2013])

Assume (A1)–(A3) and the BRW to be non-lattice. Then,

- D_n converges almost surely to some D_∞ , with $D_\infty > 0$ on non-extinction,*
- there exists a constant $C^* > 0$ such that, for any $x \in \mathbb{R}$,*

$$\mathbb{P}(M_n > -(3/2) \log n + x) \xrightarrow{n \rightarrow \infty} 1 - \mathbb{E}[e^{-C^* D_\infty e^{-x}}].$$

In other words,

$$M_n + (3/2) \log n \xrightarrow[n \rightarrow \infty]{(d)} \text{standard Gumbel} + \log(C^* D_\infty).$$

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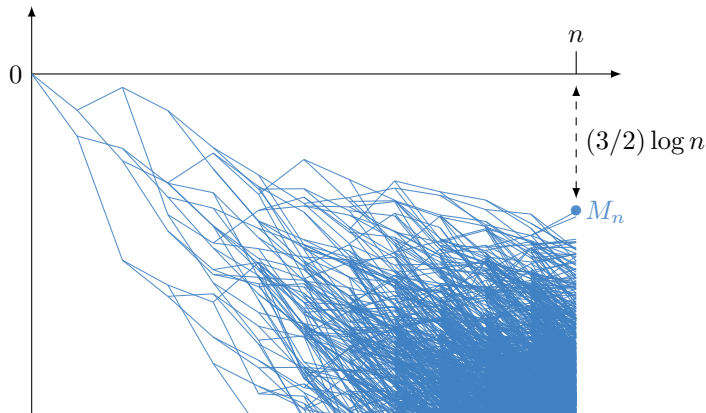


Figure: Under mild conditions, $M_n + (3/2) \log n$ converges in law as $n \rightarrow \infty$.

Maximum of the BRW: deviations

Some related papers.

Upper large deviations: $\mathbb{P}(M_n > xn)$, for $x > 0$.

- [Gantert and Höfelsauer, 2018]: large deviation principle,
- [Luo, 2025]: asymptotic equivalent.

Lower large deviations: $\mathbb{P}(M_n < -xn)$, for $x > 0$.

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Lower moderate deviations: $\mathbb{P}(M_n < -(3/2) \log n - x_n)$, for $1 \ll x_n \ll n$.

- [Hu, 2016]: moderate deviation principle when $x_n \ll \log n$,
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Upper moderate deviations: $\mathbb{P}(M_n > -(3/2) \log n + x_n)$, for $1 \ll x_n \ll n$.

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Main result

Theorem (C. and Luo, 2026+)

Assume (A1)–(A3) and consider any $(x_n)_{n \geq 1}$ such that $1 \ll x_n = O(\sqrt{n})$.

❶ If the BRW is non-lattice, then

$$\mathbb{P}(M_n > x_n - (3/2) \log n) \sim C^* x_n e^{-x_n - x_n^2/(2n\sigma^2)},$$

where

$$C^* = \lim_{\ell \rightarrow \infty} \sqrt{\frac{2}{\pi}} \frac{1}{\sigma^3} \int_0^\infty y e^y \mathbb{P}(M_\ell > y) dy \in (0, \infty).$$

❷ If the BRW is (a, h) -lattice and $x_n - (3/2) \log n \in an + h\mathbb{Z}$ for each n , then

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Upper bound: a first attempt

By the union bound and the many-to-one lemma,

$$\begin{aligned}\mathbb{P}(M_n > -(3/2) \log n + x_n) &\leq \mathbb{E}\left[\sum_{|u|=n} \mathbb{1}_{\{V(u) > -(3/2) \log n + x_n\}}\right] \\ &= \mathbb{E}[e^{-S_n} \mathbb{1}_{\{S_n > -(3/2) \log n + x_n\}}],\end{aligned}$$

where S_n is a centered random walk with variance $\sigma^2 < \infty$. Then,

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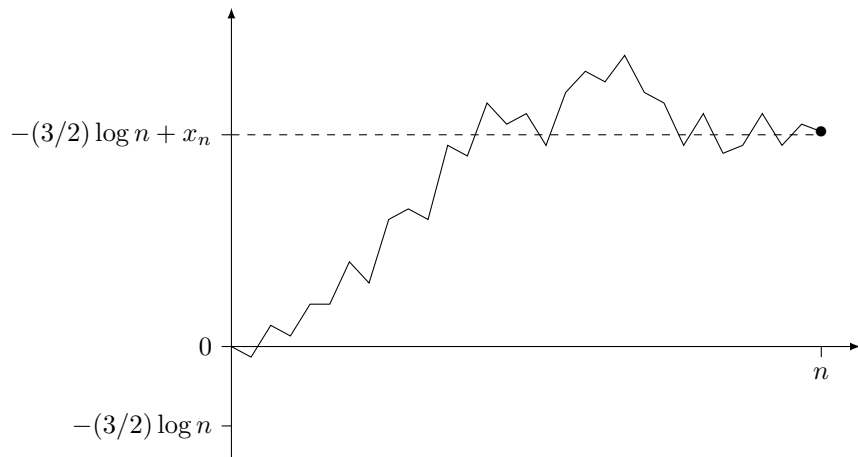


Figure: Particles performing a moderate deviation at time n come from the same ancestor at time $n - \ell$, for some large ℓ . Their previous ancestors cannot be too high.

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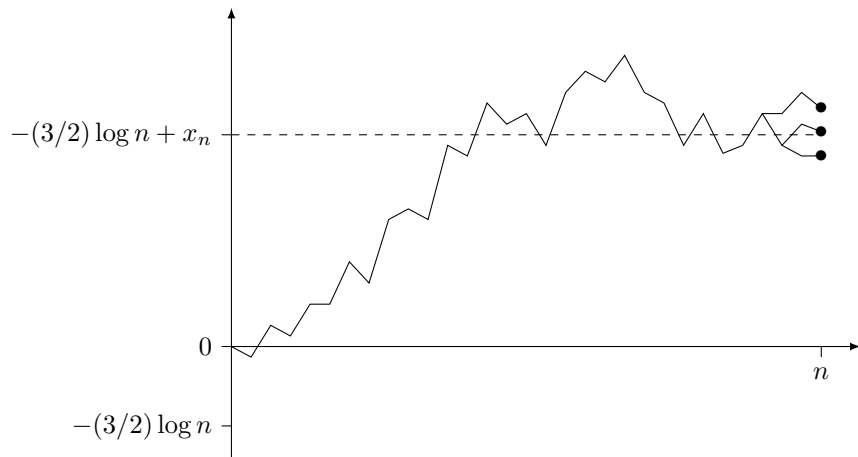


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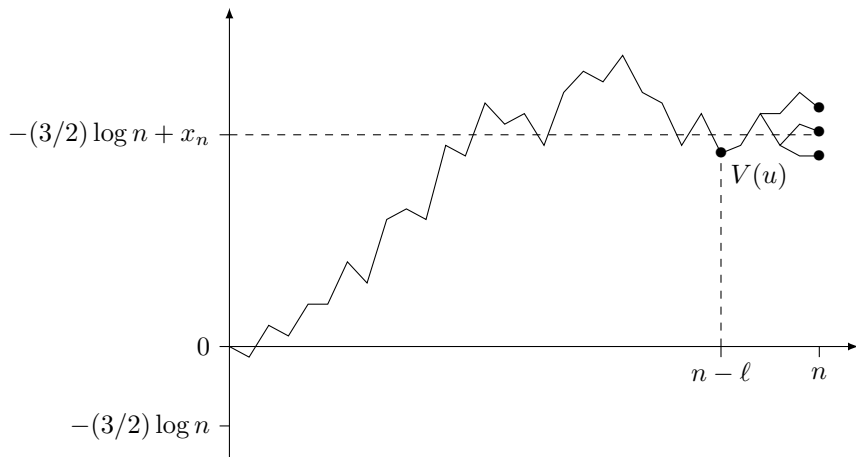


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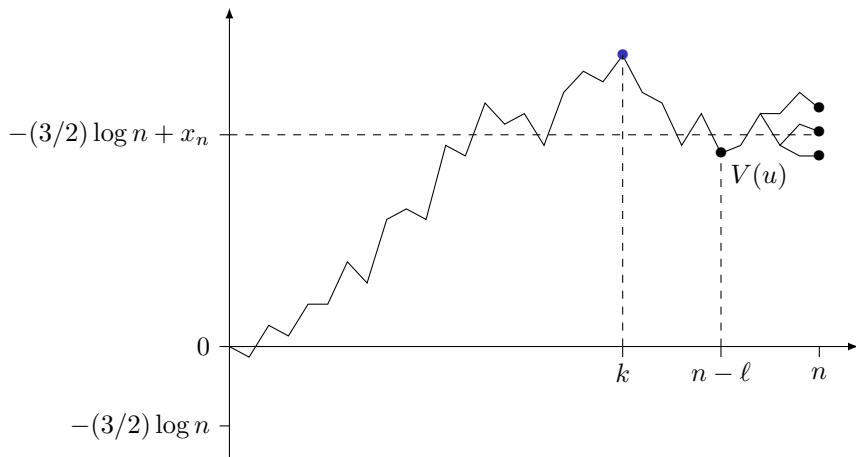


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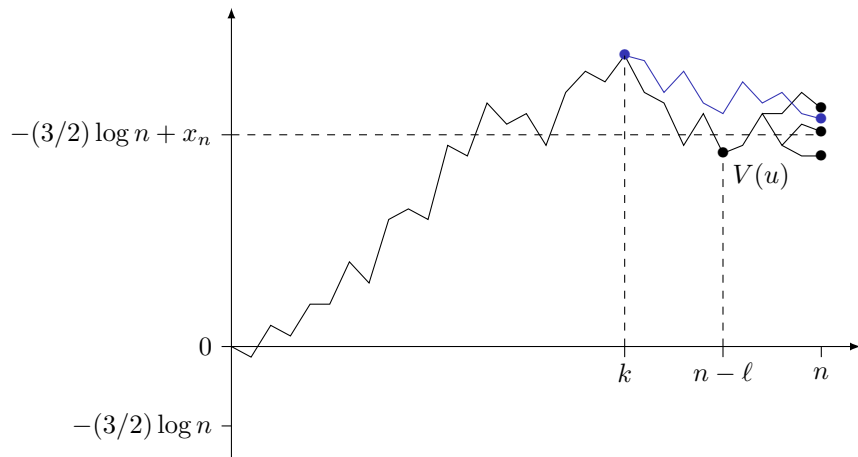


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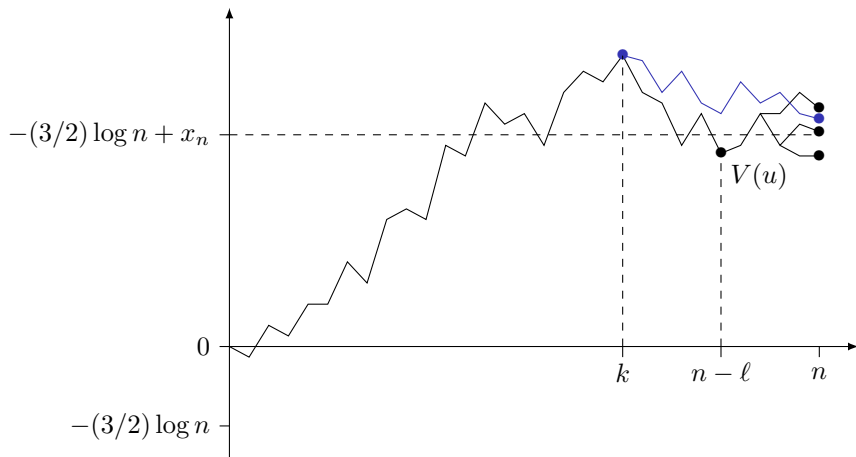


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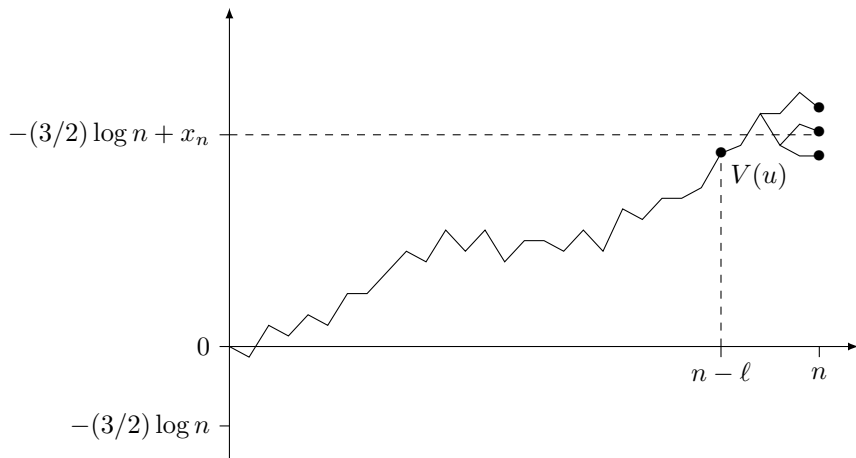


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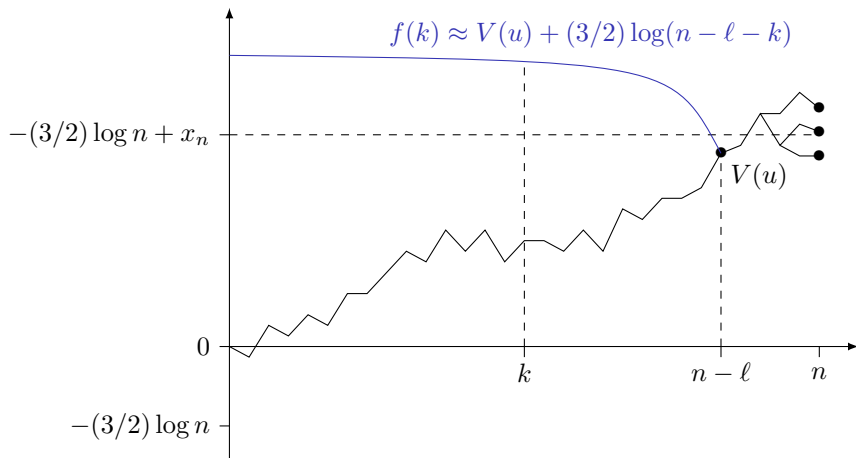


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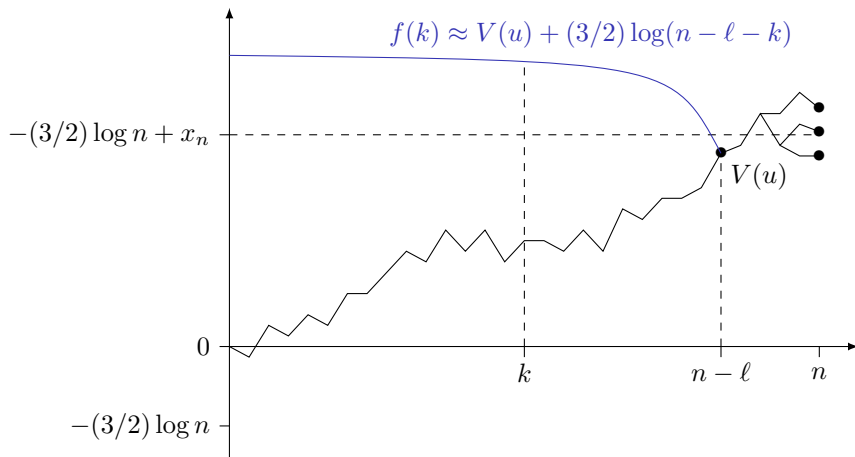


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Applications

- ④ Ancestor localization for extremal particles: For any $\varepsilon, \lambda \in (0, 1)$ and $x \in \mathbb{R}$, there exist $0 < \delta < \Delta < \infty$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\exists |u| = n, V(u_{\lambda n}) \notin [-\Delta\sqrt{n}, -\delta\sqrt{n}], V(u) > -(3/2)\log n + x) \leq \varepsilon.$$

- ⑤ Maximum of a two-speed BRW: Consider $\lambda \in (0, 1)$ and a BRW with reproduction law $\mathcal{L}^{(1)}$ until time λn and $\mathcal{L}^{(2)}$ afterwards. If $\mathcal{L}^{(1)}$ and $\mathcal{L}^{(2)}$ both satisfy (A1)–(A3) and if $\mathcal{L}^{(2)}$ is non-lattice, then

$$M_n + (3/2)\log n \xrightarrow[n \rightarrow \infty]{(d)} \text{standard Gumbel} + \log(C^{(1,2)} D_{\infty}^{(1)}).$$

- ⑥ Limit of the extremal process: By [Madaule, 2016], the extremal process converges to a decorated Poisson point process. We prove that the decoration law is the limit of the extremal process conditionally on $M_n > -(3/2)\log n + x_n$, for any $1 \ll x_n = O(\sqrt{n})$.

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- ① Ancestor localization for extremal particles: For any $\varepsilon, \lambda \in (0, 1)$ and $x \in \mathbb{R}$, there exist $0 < \delta < \Delta < \infty$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\exists |u| = n, V(u_{\lambda n}) \notin [-\Delta\sqrt{n}, -\delta\sqrt{n}], V(u) > -(3/2)\log n + x) \leq \varepsilon.$$

- ② Maximum of a two-speed BRW: Consider $\lambda \in (0, 1)$ and a BRW with reproduction law $\mathcal{L}^{(1)}$ until time λn and $\mathcal{L}^{(2)}$ afterwards. If $\mathcal{L}^{(1)}$ and $\mathcal{L}^{(2)}$ both satisfy (A1)–(A3) and if $\mathcal{L}^{(2)}$ is non-lattice, then

$$M_n + (3/2)\log n \xrightarrow[n \rightarrow \infty]{(d)} \text{standard Gumbel} + \log(C^{(1,2)} D_\infty^{(1)}).$$

- ③ Limit of the extremal process: By [Madaule, 2016], the extremal process converges to a decorated Poisson point process. We prove that the decoration law is the limit of the extremal process conditionally on $M_n > -(3/2)\log n + x_n$, for any $1 \ll x_n = O(\sqrt{n})$.

Thank you for listening!

First upper and lower bounds

Lemma ([Hu, 2016])

Assume (A1)–(A3). Then, there exists $C = C(\sigma)$ such that, for any $n \geq 1$ and $x \geq 0$,

$$\mathbb{P}(M_n > -(3/2) \log n + x) \leq C(1+x)e^{-x}.$$

Lemma ([Mallein, 2016])

Assume (A1)–(A3) and fix $A > 0$. Then, there exists $c = c(\sigma, A)$ such that, for any $n \geq 1$ and $x \in [0, A\sqrt{n}]$,

$$\mathbb{P}(M_n > -(3/2) \log n + x) \geq c(1+x)e^{-x}.$$

Step 1: Reduction to an expectation

Let $\ell = \ell_n$ such that $1 \ll \ell \ll n$ and consider the barrier

$$f(k) := x_n - (3/2) \log(n - \ell) + (3/2) \log_+(n - \ell - k).$$

To estimate $\mathbb{P}(M_n > -(3/2) \log n + x_n)$, we count particles at time $n - \ell$ such that:

- a descendant is above $-(3/2) \log n + x_n$ at time n ,
- the ancestors trajectory stays below the barrier $f(k)$ until time $n - \ell$.

In other words, we consider

$$\Lambda_{n,\ell} := \sum_{|u|=n-\ell} \mathbb{1}_{\{\max_{v \succ u} V(v) > -(3/2) \log n + x_n\}} \mathbb{1}_{\{\forall k \leq n-\ell, V(u_k) \leq f(k)\}}.$$

Strategy:

- 1 Show that $\mathbb{P}(M_n > -(3/2) \log n + x_n) \sim \mathbb{E} \Lambda_{n,\ell}$.
- 2 Show that $\mathbb{E} \Lambda_{n,\ell} \sim C^* x_n e^{-x_n - x_n^2/(2n\sigma^2)}$.

Step 1: Reduction to an expectation

Lemma 1

Assume (A1)–(A3). Then, for any choice of $1 \ll \ell \ll n$, we have $\mathbb{P}(M_n > -(3/2) \log n + x_n) \sim \mathbb{E}\Lambda_{n,\ell}$.

Proof.

④ **Upper bound:** By the union bound,

$$\mathbb{P}(M_n > -(3/2) \log n + x_n) \leq \mathbb{E}\Lambda_{n,\ell} + \mathbb{P}(G_{n,\ell}),$$

where $G_{n,\ell} = \{\text{a particle alive at time } n - \ell \text{ has a descendant above } -(3/2) \log n + x_n \text{ at time } n \text{ and one of its ancestors above } f(k) \text{ before } n - \ell\}$.

- Using [Hu, 2016], we show that $\mathbb{P}(G_{n,\ell}) \ll x_n e^{-x_n}$.
- By [Mallein, 2016], $\mathbb{P}(M_n > -(3/2) \log n + x_n) \geq c x_n e^{-x_n}$.

Hence,

$$\mathbb{P}(M_n > -(3/2) \log n + x_n) \leq (1 + o(1)) \mathbb{E}\Lambda_{n,\ell}.$$

Step 1: Reduction to an expectation

② **Lower bound:** By Cauchy–Schwarz,

$$\mathbb{P}(M_n > -(3/2) \log n + x_n) \geq \mathbb{P}(\Lambda_{n,\ell} \neq 0) \geq \frac{(\mathbb{E}\Lambda_{n,\ell})^2}{\mathbb{E}(\Lambda_{n,\ell}^2)}.$$

It would then suffice to show that $\mathbb{E}(\Lambda_{n,\ell}^2) \sim \mathbb{E}\Lambda_{n,\ell}$.

Problem: $\mathbb{E}(\Lambda_{n,\ell}^2)$ is too large in general...

Solution: Consider K large and approximate $\Lambda_{n,\ell}$ with a restriction $\Lambda_{n,\ell,K}$ to particles that do not have “too many children too high”.

- Under (A3), $\lim_{K \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{E}\Lambda_{n,\ell} / \mathbb{E}\Lambda_{n,\ell,K} = 1$.
- For K large enough, $\mathbb{E}(\Lambda_{n,\ell,K}^2) \sim \mathbb{E}\Lambda_{n,\ell,K}$ as $n \rightarrow \infty$.

We deduce that

$$\mathbb{P}(M_n > -(3/2) \log n + x_n) \geq (1 + o(1)) \mathbb{E}\Lambda_{n,\ell}.$$



Step 2: Proof of the equivalent

Lemma 2

Assume (A1)–(A3) and the BRW to be non-lattice. Then, for any choice of $1 \ll \ell \ll n$, we have $\mathbb{E}\Lambda_{n,\ell} \sim C^* x_n e^{-x_n - x_n^2/(2n\sigma^2)}$.

Proof.

- 1 By the branching property and the many-to-one,

$$\begin{aligned}\mathbb{E}\Lambda_{n,\ell} &= \mathbb{E} e^{-S_{n-\ell}} \mathbb{1}_{\{\forall k \leq n-\ell, S_k \leq f(k)\}} \mathbb{P}_{S_{n-\ell}}(M_\ell > -(3/2) \log n + x_n) \\ &= \int_0^\infty \mathbb{E} e^{-(f(n-\ell)-y)} \mathbb{1}_{\left\{ \begin{array}{l} S_{n-\ell} \in f(n-\ell) - dy, \\ \forall k \leq n-\ell, S_k \leq f(k) \end{array} \right\}} \mathbb{P}_{f(n-\ell)-y}(M_\ell > -(3/2) \log n - \dots)\end{aligned}$$

We then need to compute

$$\mathbb{P}(S_{n-\ell} \in f(n-\ell) - dy \text{ and, } \forall k \leq n-\ell, S_k \leq f(k))$$

Step 2: Proof of the equivalent

By [Doney, 2012],

$$\mathbb{P}(S_n \in x - dy \text{ and, } \forall k \leq n, S_k \leq x) \sim \sqrt{\frac{2}{\pi}} \frac{xy \, dy}{\sigma^3 n^{3/2}} e^{-x^2/(2n\sigma^2)},$$

uniformly for $1 \ll x = O(\sqrt{n})$ and $1 \ll y \ll \sqrt{n}$.

We use this estimate to show that

$$\mathbb{E}\Lambda_{n,\ell} \sim x_n e^{-x_n - x_n^2/(2n\sigma^2)} \underbrace{\sqrt{\frac{2}{\pi}} \frac{1}{\sigma^3} \int_0^\infty y e^y \mathbb{P}(M_\ell > y) \, dy}_{\varphi(\ell)}.$$

2 By Lemma 1, for any choices of $1 \ll \ell, \ell' \ll n$,

$$\varphi(\ell) \sim \frac{\mathbb{P}(M_n > -(3/2) \log n + x_n)}{x_n e^{-x_n - x_n^2/(2n\sigma^2)}} \sim \varphi(\ell').$$

This implies that $\varphi(\ell)$ converges in $(0, \infty)$.

The regime $\sqrt{n} \ll x_n \ll n$

If $\sqrt{n} \ll x \ll n$, then we do not have

$$\mathbb{P}(S_n \in x - dy \text{ and, } \forall k \leq n, S_k \leq x) \sim \sqrt{\frac{2}{\pi}} \frac{xy \, dy}{\sigma^3 n^{3/2}} e^{-x^2/(2n\sigma^2)},$$

in general. We might work with a higher assumption than (A3), e.g.

$$(A2') \quad \exists \varepsilon > 0, \mathbb{E} e^{\varepsilon |S_1|} < \infty.$$

By [Stone, 1967], under (A2'),

$$\mathbb{P}(S_n \in x - dy) \sim \frac{1}{\sqrt{2\pi}\sigma} \frac{y \, dy}{\sqrt{n}} e^{-nL^*(x/n)},$$

uniformly for $x = O(n)$ and $y = O(1)$, where

$$L^*(x) = \sup_{t>0} (xt - L(t)) \quad \text{and} \quad L(t) = \log \mathbb{E} e^{tS_1}.$$

The regime $\sqrt{n} \ll x_n \ll n$

Question 1: How to deduce a local limit theorem with a barrier?

Conjecture: Under (A2'), uniformly for $1 \ll x = O(n)$ and $1 \ll y \ll \sqrt{n}$,

$$\mathbb{P}(S_n \in x - dy \text{ and, } \forall k \leq n, S_k \leq x) \sim \frac{2}{\sqrt{\pi}} \frac{xy \, dy}{\sigma^3 n^{3/2}} e^{-nL^*(x/n)}.$$

Question 2: Once we have obtained a local limit theorem with a barrier, how to deduce an equivalent for $\mathbb{P}(M_n > -(3/2) \log n + x_n)$?

Conjecture: Under (A2'), for any $(x_n)_{n \geq 1}$ such that $\sqrt{n} \ll x_n \ll n$,

$$\mathbb{P}(M_n > x_n) \sim C^* \frac{x_n}{n^{3/2}} e^{-x_n - nL^*(x_n/n)}.$$

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